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Modern Eddington Experiment 2024 (MEE2024)

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ABSTRACT

The goal of the Modern Eddington Experiment (MEE2024) is to make the most precise measurement ever of the gravitational deflection of stellar positions near the Sun during a total solar eclipse using ground-based optical images, and by so doing, we seek to demonstrate the nature of the $1/R$ relationship for deflected stars. Fourteen observing teams will perform the MEE2024 during the total solar eclipse crossing North America on April 8, 2024. Feasibility was shown in the 2017 total solar eclipse, with the expectation that the 2024 results will be orders of magnitude more precise. This is primarily due to faster cameras, a longer totality, and multiple data sets. A computer program is being developed to semi-automatically complete the data analysis. This program will include corrections for optical distortion and finding accurate star centroid locations near the Sun. This should result in a final deflection curve accurate enough to verify Einstein's formula.

1. Introduction

In 1919, shortly after Einstein published his Theory of Relativity ([Einstein, 1916](#)), Sir Arthur Eddington and his colleague Frank Dyson performed the first successful measurement of the Einstein Coefficient ([Dyson, Eddington, & Davidson, 1920](#)) defined as the deflection magnitude at the Sun's limb, nominally 1.75 arcseconds. They did this by capturing only seven stars on two glass plates. During the next 100 years, a number of attempts at the "Eddington Experiment" were undertaken with varying levels of success. These experiments are detailed in [Figure 1](#) below. Many were not successful because this experiment is very hard to complete for a variety of reasons. Decades ago, cumbersome equipment, the need to use glass plates to capture images, and the need to image the sky months before or after an eclipse to determine the actual locations of stars were all required. Errors arose from many factors, including setup and teardown, changing weather conditions for pre- and post-eclipse imaging, and traveling with glass plates and processing them properly. The history and more detailed descriptions of these past experiments are well presented in von Klüber's paper [1960](#).

Their goal was to verify the stellar deflections θ near the Sun, assumed to be equal to:

$$\theta = \frac{4 G M_{\odot}}{R_{\odot} c^2} \left(\frac{1}{R} \right) = 1.75'' \left(\frac{1}{R} \right) \quad (1)$$

where R is equal to the radial distance from the center of the Sun in units of solar radii. At the limb of the Sun, where $R=1$, the angular deflection of a photon is 1.75 arcseconds, which is called the Einstein Coefficient (EC). The inverse law was assumed to be correct ([Froeschlé, Mignard, & Arenou, 1997](#); [Lambert & Le Poncin-Lafitte, 2009](#); [Lambert & Le Poncin-Lafitte, 2011](#); [Mignard, 2002](#); [Shapiro, Davis, Leback & Gregory, 2004](#)) and the experimenter's goals have only been to measure the EC as accurately as possible.

HISTORICAL PERFORMANCES

<u>ECLIPSE</u>	<u>RESULT</u>
1914 Crimea	Failed due to outbreak of WWI
1918 Goldendale Wash	Confused, maybe 0.87"
1922 Australia	$1.72'' \pm 0.11''$, $1.74'' \pm 0.3''$, $1.77'' \pm 0.3''$
1926 Sumatra	Failed
1929 Sumatra	$1.75'' \pm 0.13''$
1936 Russia, Japan	$1.70'' \pm 0.40''$, $2.73'' \pm 0.31''$
1947 Brazil	$2.01'' \pm 0.27''$
1952 Khartoum	$1.70'' \pm 0.10''$
1973 Mauritania	$1.66'' \pm 0.18''$



Figure 1
Historical Performances of the Eddington Experiment.

The attempt in 1922 solidified the results from 1919 by obtaining more stellar images, shown in [Figure 2](#). Although having more data was convincing, there were no stars closer than two solar radii, and the correct formula could not be demonstrated from the curve. Subsequent experiments by van Biesbroeck [1953](#) and Jones [1976](#) had the same problem; few stars close to the Sun and a wide scatter of deflections for stars at larger radii. After 1973, there were no known experiments performed for over four decades.

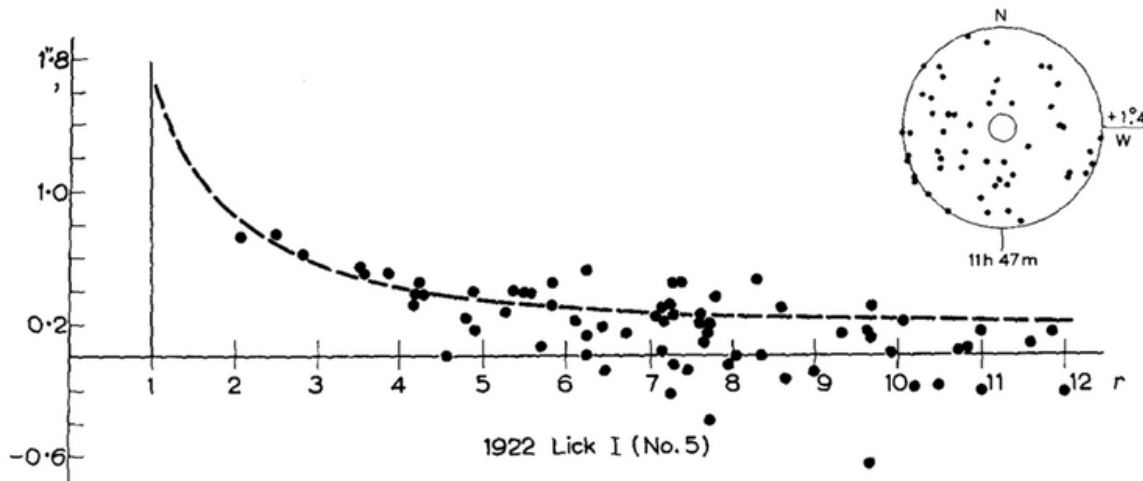


Figure 2

A plot of the results from the 1922 eclipse reproduced from von Klüber (1960).

In 2017 the era of the Modern Eddington Experiment began, mainly because of computer-controlled equipment and electronic cameras. Another key element in the modern experiment is the use of the extremely accurate stellar positions from the Gaia satellite; taking comparison images months earlier or later is no longer required. The data processing procedure is still very complex. Therefore, a secondary goal of this project is to make the overall experiment accessible for future amateur astronomers, faculty, and students in upper-level lab courses in the college curriculums. The creation of a Modern Eddington Experiment Lab Manual and computer software for station control during data collection and post eclipse data processing is essential to meet this secondary goal. In the long course of time, this secondary goal could prove to be more important to the mission of MEE2024 than the primary goal of verification of the deflection power law.

In 2015, planning began to repeat this experiment during the total solar eclipse that crossed the United States in 2017. The equipment set up in Wyoming, along with the final results, is shown in [Figure 3](#). For this eclipse, two bright stars just happened to be located about $1.5 R_{\odot}$ from the Sun, a coincidence that will not be repeated for decades. This data set was good enough to derive a very precise Einstein Coefficient of 1.751 arcsec, when the theoretical value is also 1.751 arcsec. The uncertainty of 3% came from atmospheric turbulence ([Bruns, 2018](#)), after assuming an inverse power law. Up to the present, every experiment performed with optical and radio telescopes, both on the ground and satellites assumed the hyperbolic $1/R$ power law. To adequately verify the inverse power law, many additional stellar positions measurements will be necessary, especially in the region of the corona from $R = 1$ -2.



Figure 3
Telescope setup for experiment for the 2017 total solar eclipse.

The data analysis plan applied to the data displayed in Figure 2 used the exponent as a variable in a least-squares fit. An analysis of the Bruns 2017 data ([Bruns, 2018](#)) resulted in the formula:

$$\theta = \frac{1.96}{R^{1.13}}$$

When the exponent was constrained to -1, the result was:

$$\theta = \frac{1.751}{R}$$

a very nice fit to Einstein's theory. Each data was the result of a single measurement and had no easily defined error associated with it. The radial distance from the theoretical curve is the only measure of the error.

Individual measured centroids might be analyzed to have an error proportional to the FWHM and SNR of each point, but that varied too much, due to the variable solar background, to have much meaning. Those errors ranged from 0.1 arcsec to 0.01 arcsec. The final curve did not use a weighting factor based on these individual errors; each point was weighted equally.

This same procedure will be applied to the 2024 data sets, with the expectation that the best fit will be close to $1/R$ and the coefficient close to 1.75. The residual errors, after subtracting the $1.75/R$ baseline, will be examined for possible fits to higher powers. If there is enough good data to show a good correlation, the results will be offered to other interested experimental and gravitational experts.

Another Modern Eddington Experiment was performed in Oregon in 2017 by Berry and Dittrich using similar equipment as the Bruns experiment. While the results of this experiment were comparable to past experiments (1.68 arcsec), four students from Portland Community College participated and became the first students in history to measure the curvature of space in the optical. The uncertainty caused by plate scale difficulties was very large, so that the results were not publishable.

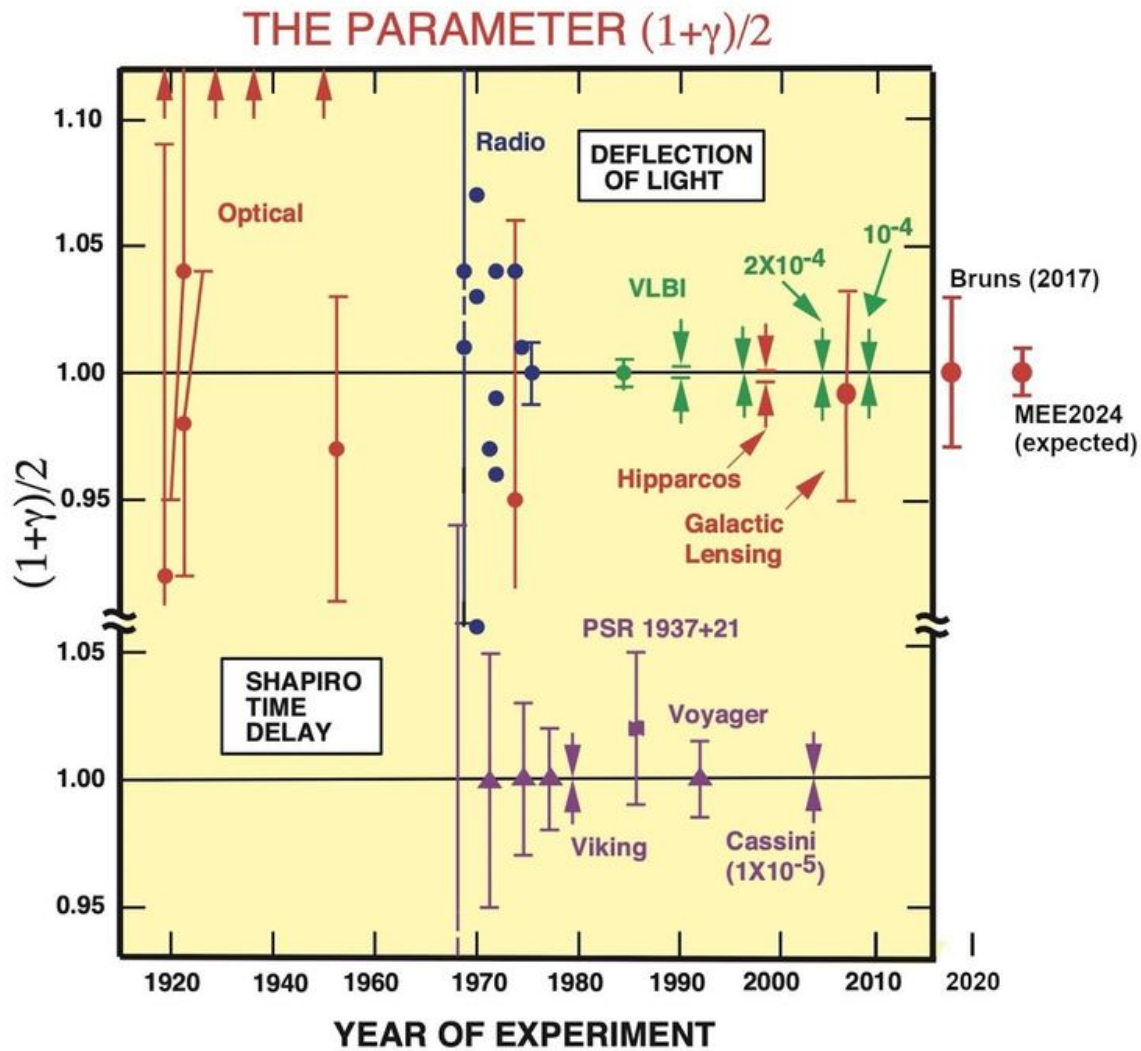


Figure 4

A plot showing the historical progression of the measurement of the deflection coefficient. Modified from Figure 2 of Will (2015).

Figure 4 shows the historical experiments in the optical and the experiments undertaken by single radio telescopes, Very Long Baseline Interferometry of radio telescopes (VLBI) and the Hipparchus satellite. This figure is modified from Figure 2 of Will (2015). The result is displayed as $A = 1.0$ if the deflection at the limb of the sun is 1.75 arcsec, while the error bars are shown by size.

In 2024, with exposure times as fast as 100msec with zero readout delay, the fastest CMOS cameras will allow us to capture stars closer to the Sun's limb. The larger format cameras are slower but capture stars out to a larger radius. With 10 telescopes, this will give us 10 final-processed images with the likelihood of measuring thousands of stars. Deflections of each star may be measured to a 1% accuracy, based on current tests. The resulting Einstein Coefficient measurement error may be of order 0.001 arcseconds (0.1%), which would be the

most accurate measurement performed in the optical range. This result has been added to [Figure 4](#) on the right. The predicted uncertainty is based on acquiring data from 10 telescopes [10x], using more efficient cameras [8x], wider fields of view [2x] a longer eclipse [2x], and less atmospheric turbulence [2x]. This leads to a factor of $\sqrt{10 \times 8 \times 2 \times 2} = 25x$ improvement over the 3% error from 2017. The large range of stars would finally allow a precise calculation of the deflection power law.

Now, we have put together 14 observing teams of physicists, astronomers, amateur observers, and students to perform the Modern Eddington Experiment (MEE2024) during the total solar eclipse crossing North America on April 8, 2024. This project has two main goals:

- Demonstrate the inverse power law of the General Theory of Relativity when applied to stellar deflections measured near the Sun.
- Produce the best determination of the deflection coefficient of stellar positions due to the Sun's gravity using optical images from the ground.

Colleges and universities worldwide use an Advanced Lab course in the senior curriculum for physics/astronomy majors and yet there is generally no experiment in this curriculum that involves General Relativity (GR). It is important to have such an experiment for undergraduates because GR and the curvature of space takes such an important and popular place in physics and astronomy today. Other experiments to involve students in GR are complicated and involve expensive equipment that most schools do not have. The Modern Eddington Experiment, with our completed lab manual and open-source data processing software, will facilitate the future performance of the experiment by student and amateur astronomers during the eclipse in 2027 and beyond.

An exciting aspect of this project is its members. The educational institutions involved are from around the world and include community colleges and primarily undergraduate institutions. In addition, we have advanced amateur astronomers involved. With the benefit of their experience and expertise in setting up telescopes and acquiring quality data, our groups expect to succeed with excellent results.

2. Data & Methodology

One limiting factor in the experiment is the atmospheric turbulence that moves the stars randomly around the image. This can only be reduced by integrating over a longer period. The 2024 eclipse lasts twice as long as the 2017 eclipse, and the CMOS cameras have no dead time in the acquisition process as there was with CCD cameras. The CMOS cameras will thus integrate the turbulence effects over eight times longer than in 2017. Additionally, the 2024 eclipse occurs higher in altitude above the horizon, further reducing turbulence. These factors will significantly reduce the scatter in the measured deflections.

Some telescope-camera combinations also cover a larger field of view, so stars farther from the Sun can be measured. Hundreds of stars are expected in the 2024 images, compared to dozens in the 2017 experiment. The

field of view centered on the Sun during totality for the narrow and wide-angle CMOS cameras is shown with the actual 2024 star field in [Figure 5](#). Stars are shown with a magnitude range of 4.3 on the bright end and 12.5 on the faint end. Magnitude 11 stars are likely measurable close to the Sun, and magnitude 13 stars at the edges.

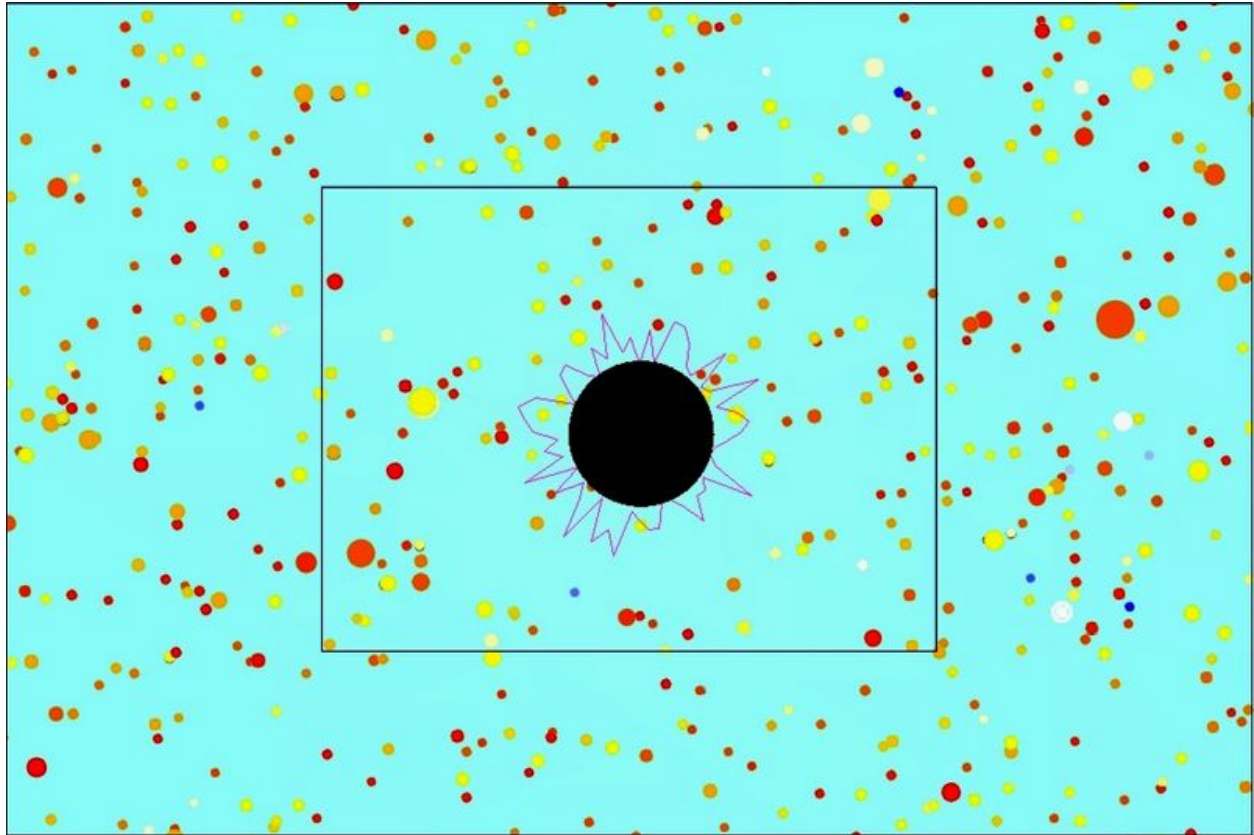


Figure 5

The 2024 eclipse field of view (FOV) centered on the Sun during totality with right ascension (RA) increasing to the left and declination (DEC) increasing up. The inner box is for a ZWO 1600 CMOS camera giving a 142 x 107 arcminute FOV, and the outer box is a wide-angle ZWO 6200 CMOS camera with a larger 289 x 193 arcminute FOV. Star colors are approximations of real star color and are only meant to be representative.

An initial test of what our results might look like is shown in [Figure 6](#), where we have simulated the deflections for the eclipse field in 2024. The telescope field of view was based on the Televue NP101 refractor with a 0.8x field flattener and a ZWO 1600 CMOS camera. This combination gives us a total of 116 stars. We used the theoretical relationship to determine the deflection for each star and ran a Monte Carlo simulation to determine how well we can recover both the deflection constant and the exponent of the power law in different circumstances. We ran 1,000 randomized simulations by adding a random Gaussian error to each stellar deflection and then fitting the resulting points to determine the deflection coefficient and power law exponent. With a typical individual position error of 0.07 arcseconds, we can recover both the coefficient and exponent to an accuracy of 5%. This demonstrates how important it is to recover as many stars as possible from the eclipse images.

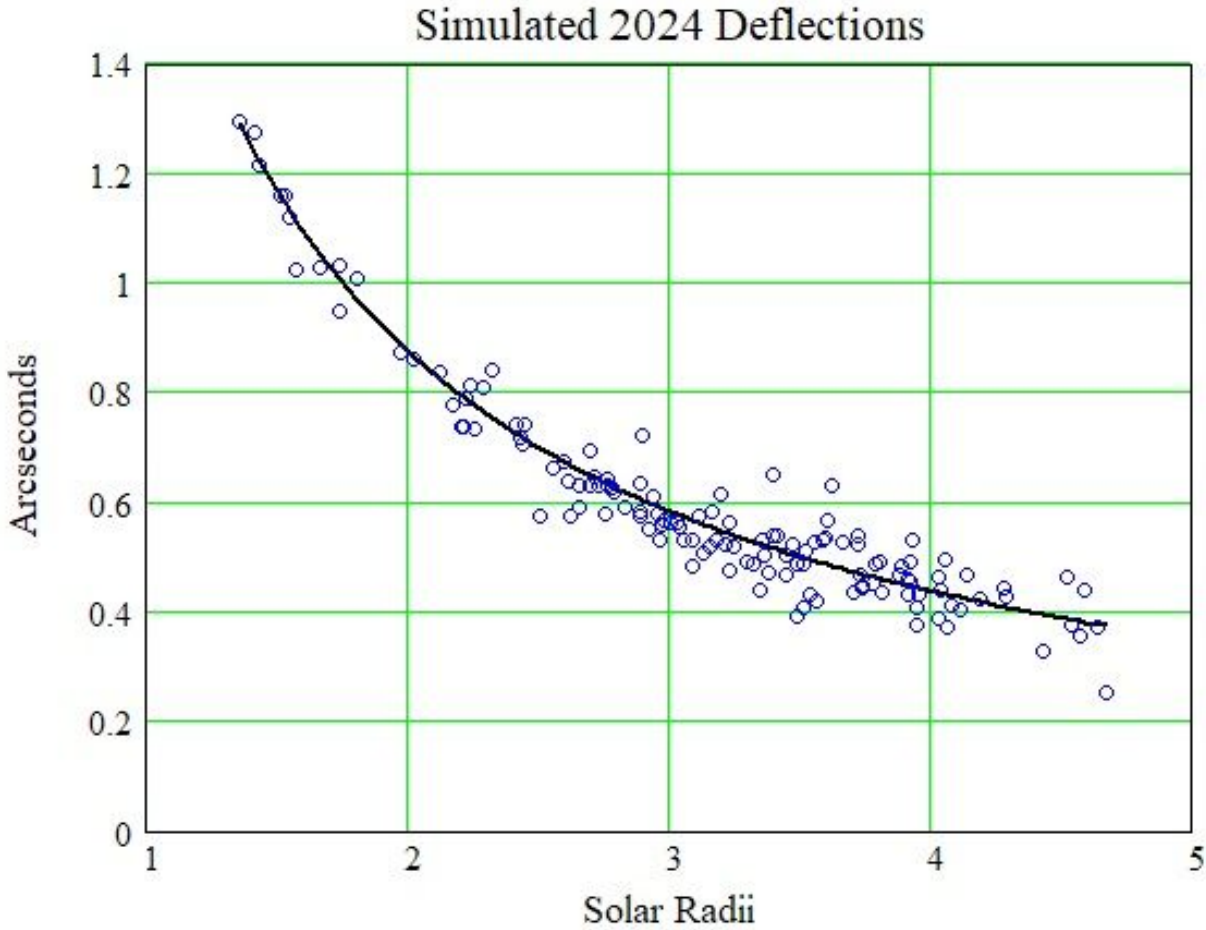


Figure 6
Simulated 2024 results for one telescope and camera combination.

With the larger format ZWO 6200 CMOS camera we will obtain several times more stars in the eclipse field. Therefore, we can expect a further reduction in the errors for both curve fit verification and the determination of the EC. If, on the other hand, we assume an inverse power law for the relationship and only fit for the deflection coefficient, our estimated error from the above simulation drops to 1%, which would be the best ever determination of the coefficient using optical ground-based observations of stars during a total eclipse.

The MEE2024 project has now been organized with 14 stations. Since each one will be set up with different telescopes, cameras, and exposures, the data sets should exhibit randomness among them. This justifies combining the results from all of the stations to reduce the data scatter even more. The final curve in this project should result in an accurate determination of the correct inverse power law.

2.1 Detailed Data Acquisition and Analysis Procedures

To get the best possible final value for the gravitation deflection coefficient, the detailed data acquisition and data analysis procedures summarized here will be followed. They are slightly modified from the manual

procedures followed in 2017 that resulted in historically precise results ([Bruns, 2018](#)). In 2024, we expect well over 100 stars surrounding the Sun from each telescope station, the result of combining thousands of images. These large numbers require new data analysis software.

The first steps are acquiring the different images and processing them so that the centroids can be determined. Analyzing those images to determine the distortion coefficients and the plate constants comes next. The final step is to determine the gravitational deflection values for each star. This procedure is presently being organized into a master software program and will be automated as much as practical. In the following we will provide more details on each of these steps, and further details of data acquisition and analysis are presented in Bruns ([2018](#)).

2.1.1 Image Acquisitions and Processing

Telescope optical distortions should be measured at night, before or after the eclipse, with as little change in the camera/telescope as possible. To get the most accurate final numbers re-focusing may be required, but disassembly should not be done. Longer exposures should be taken at night for these calibration images; the purpose is to average over turbulence to reduce centroid noise. These image series should contain thousands of measurable stars. Commercial software can be used to obtain the optical distortion coefficients.

The eclipse day imaging must occur in a window slightly less than 4.5 minutes long so the mount and camera should be fully automated. Nominally, the plate scale calibration fields would be 5 to 10 degrees distant from the Sun, where the deflections will be very small and practically uniform across the field. The first field should be imaged at the start of totality, for approximately 50 seconds. The field surrounding the Sun is then imaged during the deepest part of totality, for 160 seconds. The second field is then imaged for the final 50 seconds of totality. These calibration images will give accurate plate scales, even though the integrated exposure times are relatively short, because of the large number of stars. Variations in the procedures might include a set of four images, with the Sun placed in each corner for 60 seconds. For the largest cameras, simply centering the Sun for the total duration is an option, using stars in the corners for plate scale calibration.

Prior to determining the centroids, the individual images for each field should be aligned and stacked into master images. The thousand or more images will be automatically aligned to produce a master image. The fields containing the Sun show very few stars prior to additional processing, but only a few stars far from the Sun are enough to align the images. The next step is to remove the bright background from the solar corona so many more stars will be measurable. Early testing showed that this can be done using a blurred version of the image, but additional testing is needed with the new software.

2.1.2 Centroid Locations and Star Matching

In 2017, centroids were found using three commercial software programs: MaxIm DL, Astrometrica, and PinPoint. The centroids from each of these programs had slightly different coordinates, so a comparison of the

centroid results as determined by slightly different centroiding methods was warranted. The maximum difference between the centroids was typically less than 0.1 pixel and the standard deviation of the difference was 0.02 pixels. To reduce the errors for MEE2024, centroiding will be done in the new data processing software that has been developed. While creating this new Python program using elements from the AstroPy library, the program has similar centroiding styles. The new software has been compared favorably in performance with Astrometrica and it was built to handle larger images.

2.1.3 Capturing stars with Large Deflections

A key component of the goals of MEE2024 is to image stars close to the limb of the Sun, where the location is less than $2 R_{\odot}$ and the deflections are larger than 1 arcsecond. This requires short exposures to prevent saturating the camera from the Sun's corona. Since the CMOS cameras download while imaging, exposures as short as 0.1 sec are possible without losing integrated exposure time. With 160 seconds of totality dedicated to the images close to the Sun, capturing 1600 images per station is possible. To get stars even closer to the Sun, some stations might use 0.025 second exposures, with an integrated exposure still exceeding 40 seconds. This might show a few bright stars with large deflections very close to the Sun's limb.

2.1.4 Corrections for the Master Centroid List

Once a preliminary list of centroids with matching star coordinates has been generated, it must be edited to remove bad or potentially bad centroids. This includes centroids too close to the image borders, centroids that are abnormally broad or narrow, or centroids that contain saturated pixels. The centroid list can be plotted to visually search for any other anomalies. Close double stars might be noted, and those can be removed to avoid possible errors. An automatic program to search for all nearest-neighbor combinations, and deleting those within a few pixels, will be completed.

Corrections to the raw centroid locations must be made by the automated software, which include the effects of:

1. Optical Image distortion (typically <3 arcseconds and varies across the field),
2. Differential atmospheric refraction (typically 2 arcseconds for the entire field) which is a result of the Sun's altitude of about 70 degrees,
3. Random seeing wobbles (typically 0.5 arcseconds and it random across the field) from air turbulence,
4. Stellar proper motion and parallax (typically <0.01 arcseconds and specific to certain stars), and
5. Differential relativistic aberration (typically 0.01 arcseconds for the entire field) which is the effect of the motion of the earth and solar system.

Optical image distortion is corrected by measuring the positions of stars near the zenith during nighttime calibrations. Differences in the measured location compared to the perfect telescope give a set of coefficients up to cubic order in distance from the optical axis. This can be done before or after the eclipse date, since the

coefficients are related to the design of the optics and should remain stable. A good observing session will image several star fields near zenith, integrating for at least one minute to reduce turbulence.

Differential atmospheric refraction, stellar proper motion and parallax, and relativistic aberration will be corrected by utilizing the US Naval Observatory NOVAS program. The output of NOVAS corrects the Gaia catalog star positions to the observation date and for refraction based on the local weather conditions. This list is the precise apparent location of where every star appears in the field, to micro-arcsecond precision.

Atmospheric turbulence creates the largest effect on uncertainty even though optical image distortion is larger in size. These centroids dance around randomly with a magnitude up to an arcsecond in short exposures but averaging over even 30 seconds will reduce the variations to the 0.1 arcsecond range. Turbulence is a function of the local topography, local temperature differences, and location of the high-altitude jet stream. Since the location of the telescopes for this project are determined by cloud predictions more than anything else, there is not much that can be done to reduce this effect except integrate the exposures for as long as possible.

These corrections provide the measured coordinates of all stellar positions, which can be compared with the apparent positions from the NOVAS-corrected list. The resulting deflections and distance from the center of the Sun are then plotted. With a least-squares fit, the EC and the power law coefficient are determined. This is the same procedure that was used to produce the simulated results shown in [Figure 6](#). The data from each telescope/camera system will be evaluated, and weighting the solutions to each data set or telescope station is an option, with the largest weights attached to those data sets with the smallest uncertainties. This will result in a final deflection and power law coefficient from the 2024 eclipse that is more accurate than any previous optical determination in history. The data obtained to produce this result will reside in the open-source website www.ModernEddingtonExperiment.org.

3. Summary and Relevance to Eclipse Science

The MEE2024 project is very relevant to eclipse science. As mentioned above, this experiment can only be accomplished during a total solar eclipse. This has meant that progress on both the measurement of the deflection constant, and the nature of the deflection relationship, has been very slow. There were 60 total solar eclipses in the 20th century after Einstein's work was published in 1916, and another 14 so far in the 21st century. Experiments were not possible during all of these 74 total solar eclipses for a multitude of reasons. With typical durations of several minutes, that means there has only been a grand total of several hours that this experiment could have been performed since the prediction in 1916.

The total eclipse in 2024 will be the last to pass through the continental US for another 20 years. So, the 2024 eclipse event is a unique opportunity to perform the Eddington Experiment with minimal travel and huge impact in the US as far as outreach and community involvement are concerned.

The summary by von Klüber [1960](#) is an extensive and critical look at the history of light-deflection measurements near the Sun prior to 1960. The difficulties and problems are discussed, and the conclusion has several points and recommendations on how these experiments should proceed in the future. In his first point, he notes that the deflection of stellar positions near the Sun “quite obviously exists” and follows that up with “the observations are not sufficient to show decisively whether the deflection really follows the hyperbolic law predicted by the General Theory of Relativity”. He correctly attributes this deficiency to the small number of stars measured very close to the Sun. He also mentions that the data can “be represented quite well even by straight lines” and there were arguments at least as late as 1959 that a straight line fit better represented all the available data ([Mikhailov, 1959](#)).

One may assume, incorrectly in this case, that because these arguments and observations were made more than half a century ago, the current state of the field is very different. However, since von Klüber’s work, there have been only two successful measurements of stellar deflections during a total solar eclipse, one in 1973 ([Jones, 1976](#)) and one in 2017 ([Bruns, 2018](#)). Neither of these measurements obtained enough stars to establish the inverse power law relationship of the stellar deflections, so this remains a crucial point to be verified, more than a century after the establishment of the theory.

In his list of important points that needed to be addressed in future observations of this type, von Klüber mentions better scale values, more stars in the field, precise guiding, high- quality optics, and multiple images during the eclipse. Our MEE2024 project and resulting images will far surpass what von Klüber had in mind when he made these suggestions, and we will finally be able to verify the inverse power law of stellar deflections near the Sun, more than 60 years after this paper and more than 100 years after Einstein first made the prediction. This verification will also validate the results of the radio telescopes, VLBI efforts and now the Hipparchus and Gaia satellites – done with undergraduate students and amateur astronomers operating the telescope stations!

Appendix: MEE 2024 Participants

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Cade Freels, Astronomer – Oregon Observatory, Sun River OR

Doug Smith, Amateur Astronomer - London England

Andrew Smith, Cambridge University

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Greg Mulder, Physics Professor - Linn Benton Community College

Daniel Borrero Echeverry, Physics Professor – Willamette University

Joseph M. Izen, Professor Emeritus of Physics – University of Texas Dallas

Jesse Kinder, Physics Professor – Oregon Institute of Technology

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Dylan Villafranco, Student – Angelo State University

Stasha Youngquist, Student – Angelo State University

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